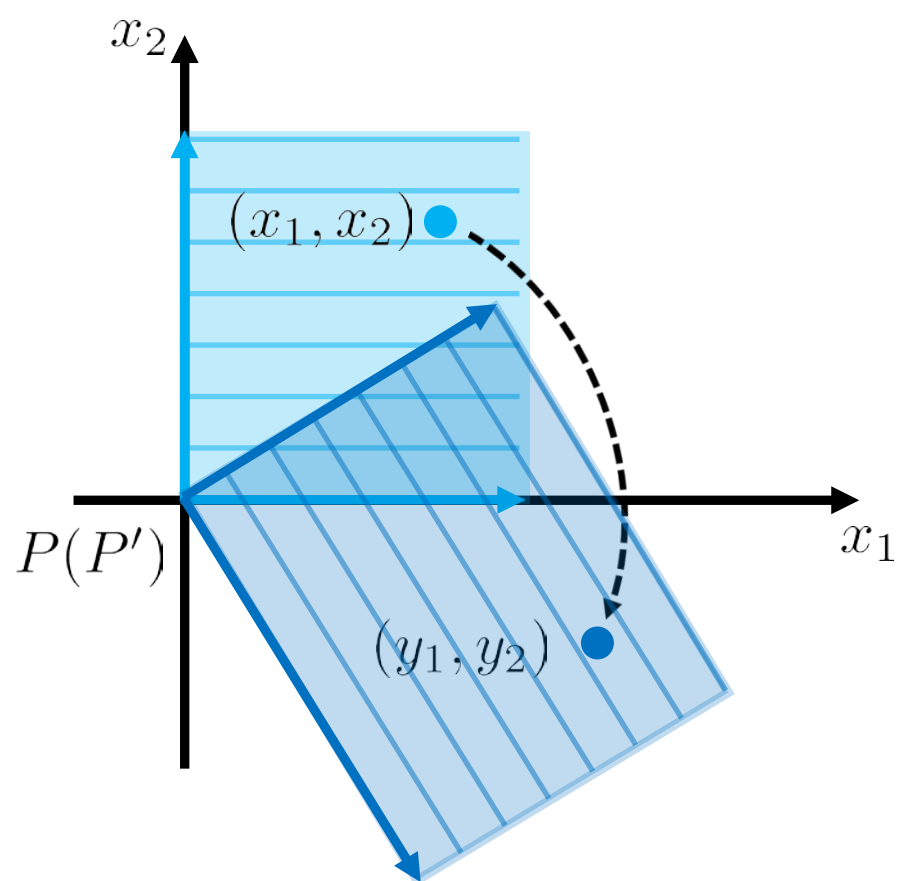
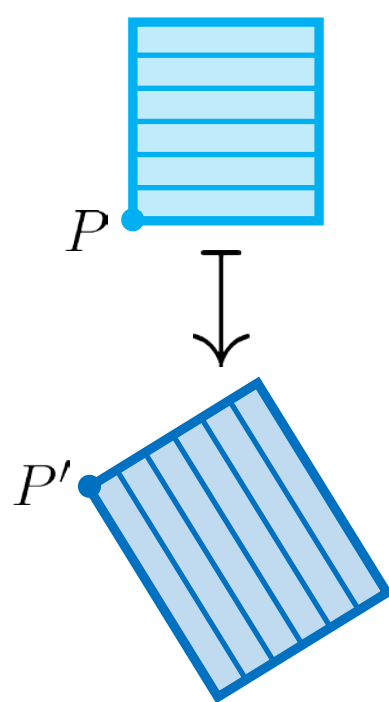
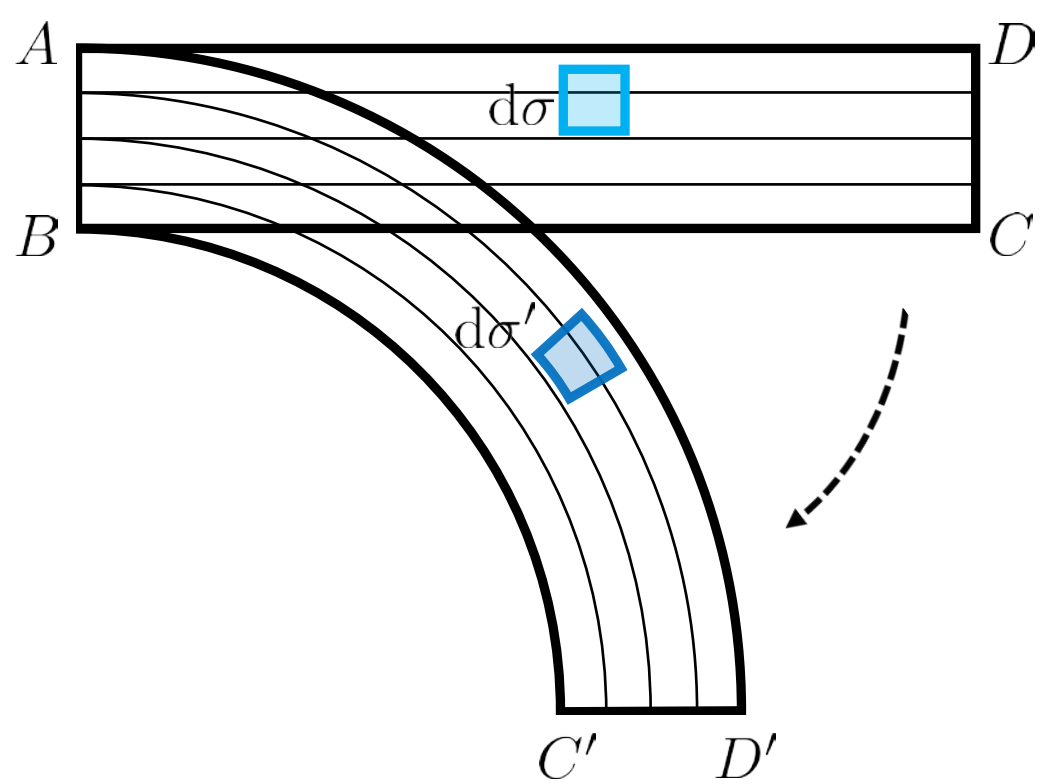
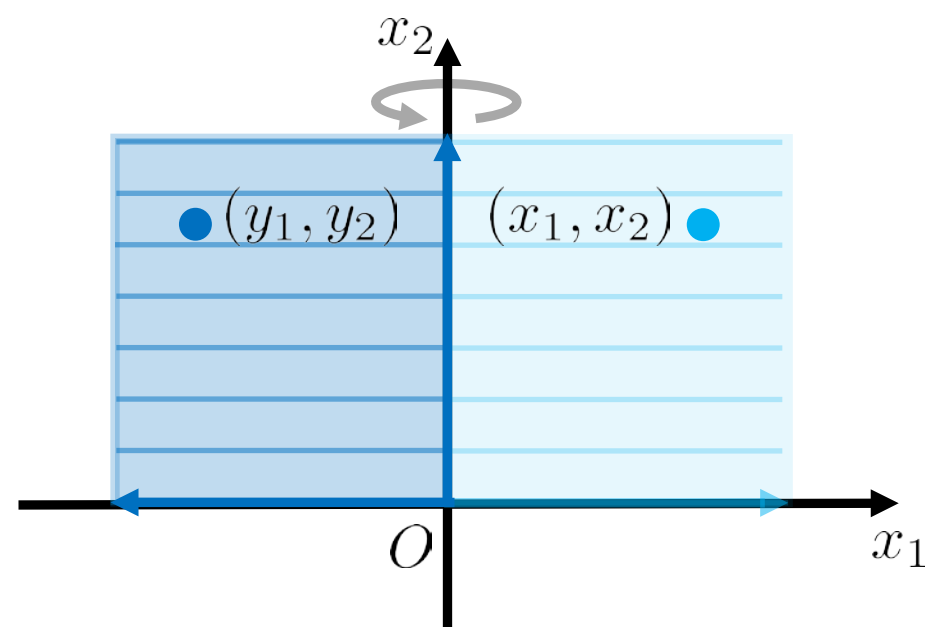
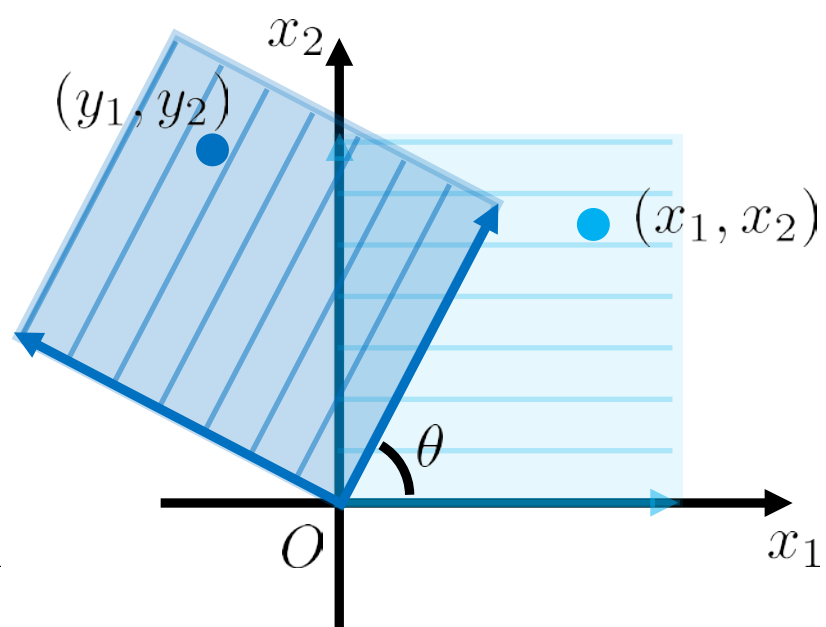
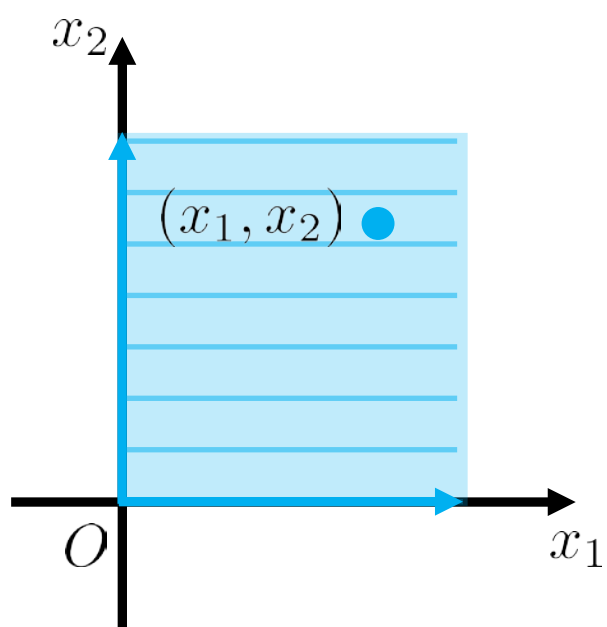
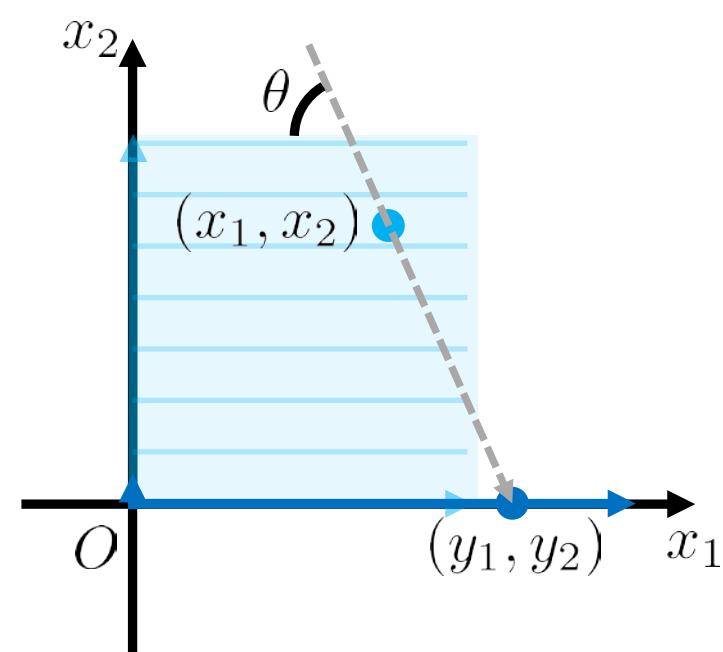
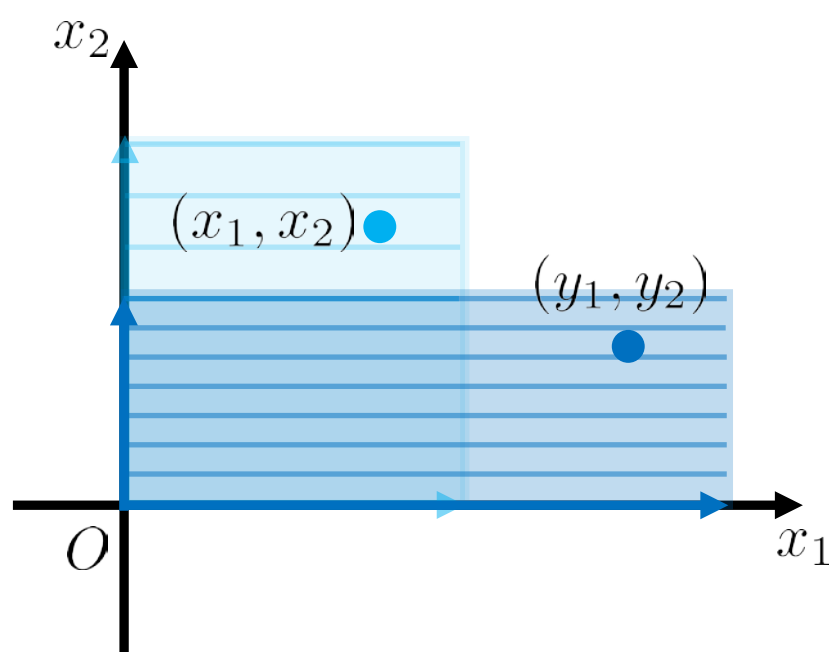
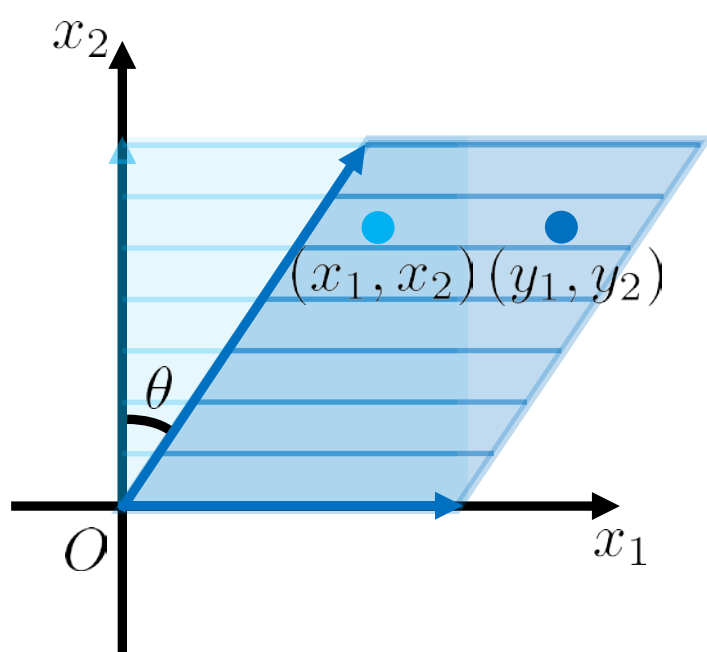






$$(1) \begin{cases} y_1 = \cos \theta \cdot x_1 - \sin \theta \cdot x_2 \\ y_2 = \sin \theta \cdot x_1 + \cos \theta \cdot x_2 \end{cases}$$

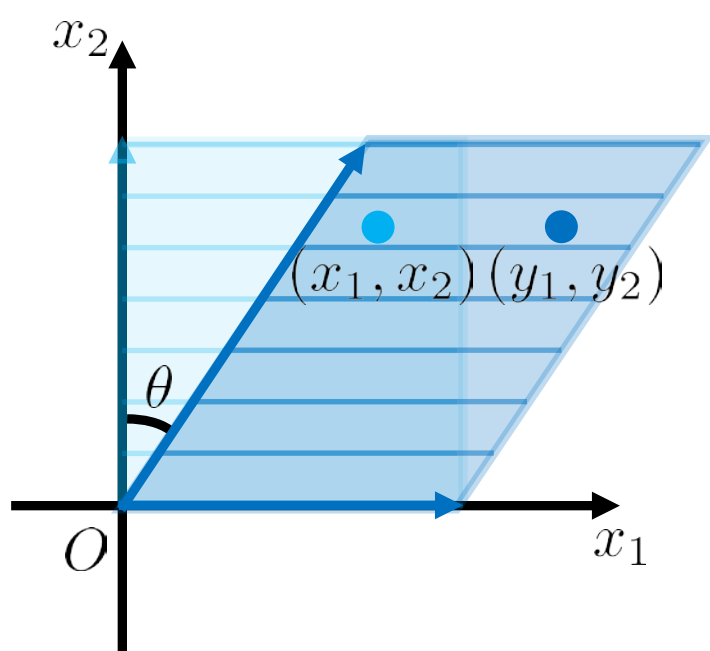
$$(2) \begin{cases} y_1 = -1 \cdot x_1 + 0 \cdot x_2 \\ y_2 = 0 \cdot x_1 + 1 \cdot x_2 \end{cases}$$



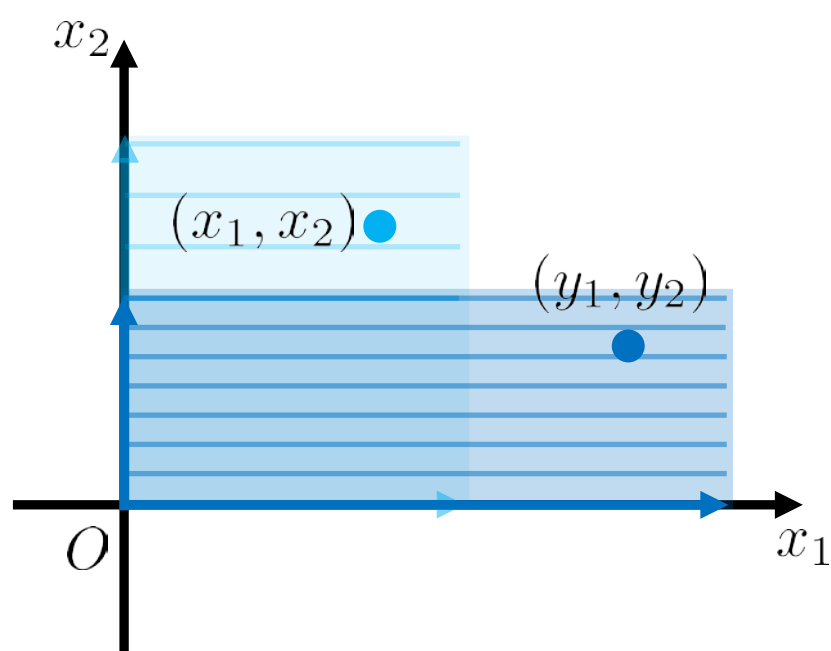
$$(3) \begin{cases} y_1 = 1 \cdot x_1 + \tan \theta \cdot x_2 \\ y_2 = 0 \cdot x_1 + 1 \cdot x_2 \end{cases}$$

$$(4) \begin{cases} y_1 = k_1 \cdot x_1 + 0 \cdot x_2 \\ y_2 = 0 \cdot x_1 + k_2 \cdot x_2 \end{cases}$$

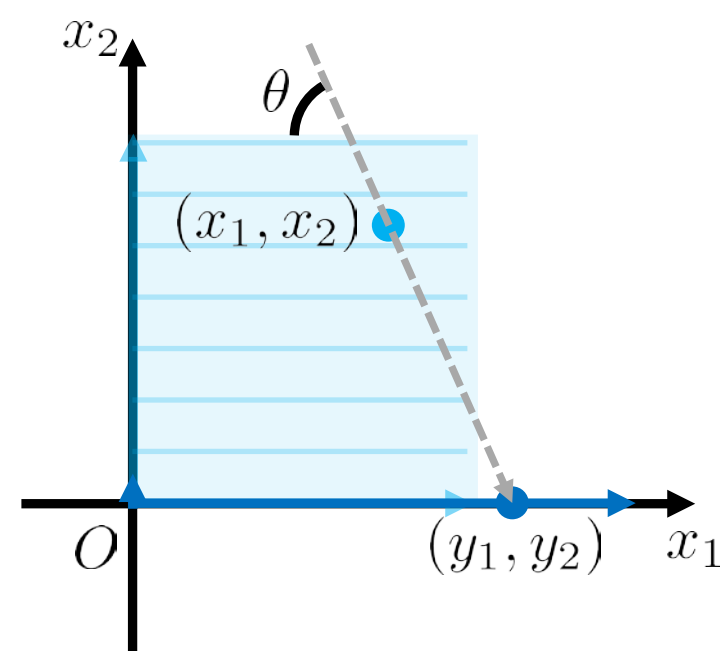
$$(5) \begin{cases} y_1 = (1 + \cot \theta) \cdot x_1 + 0 \cdot x_2 \\ y_2 = 0 \cdot x_2 + 0 \cdot x_2 \end{cases}$$



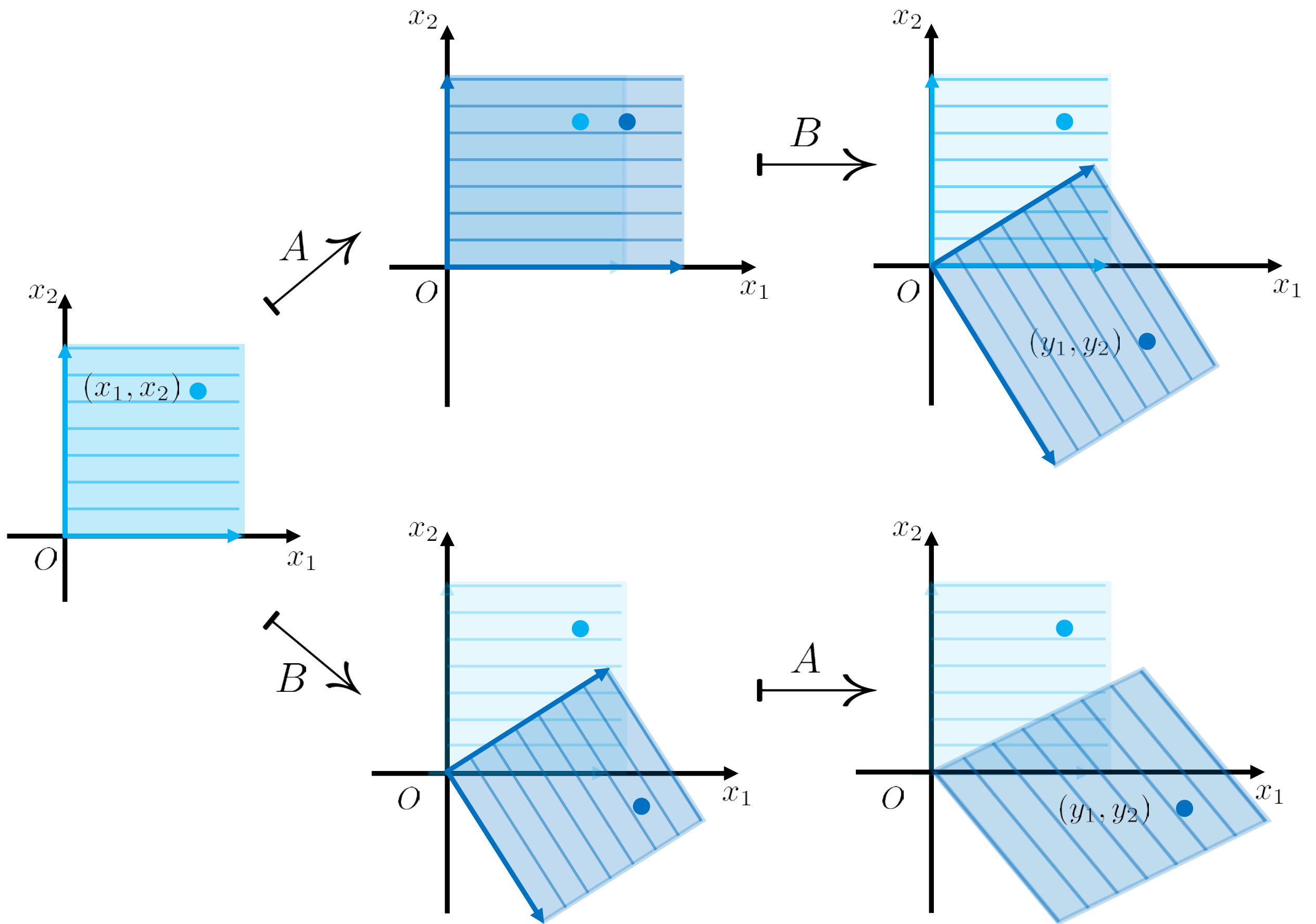
$$(3) \quad \begin{cases} y_1 = 1 \cdot x_1 + \tan \theta \cdot x_2 \\ y_2 = 0 \cdot x_1 + 1 \cdot x_2 \end{cases}$$

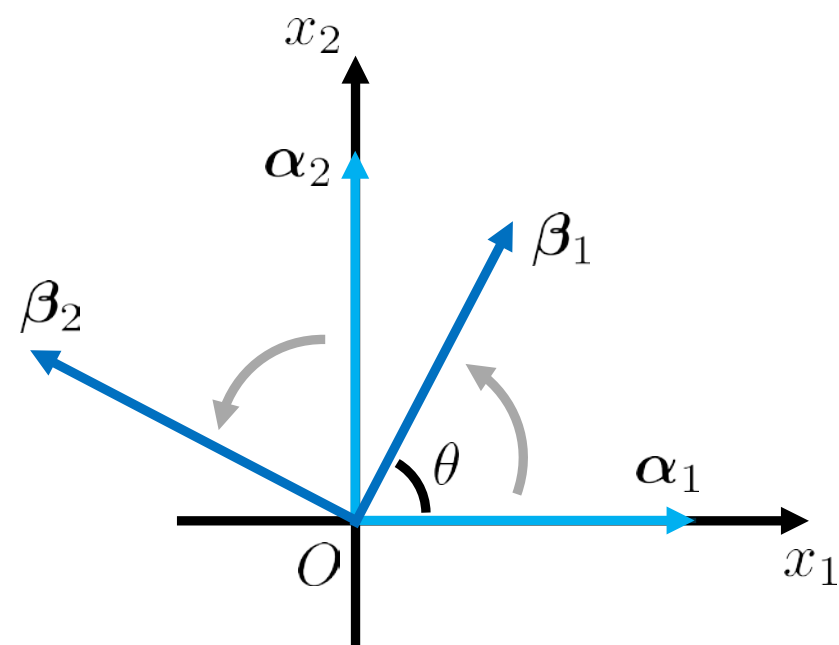
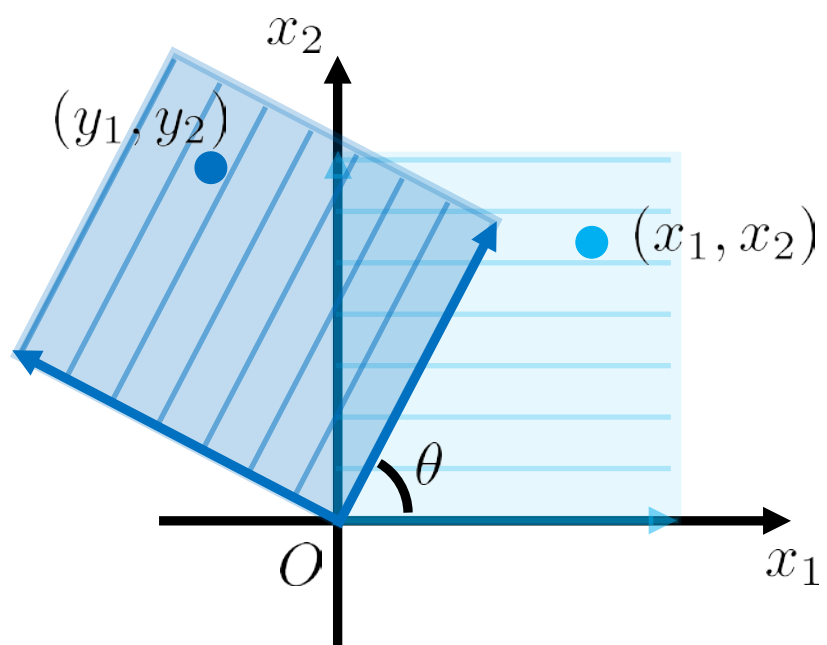


$$(4) \quad \begin{cases} y_1 = k_1 \cdot x_1 + 0 \cdot x_2 \\ y_2 = 0 \cdot x_1 + k_2 \cdot x_2 \end{cases}$$



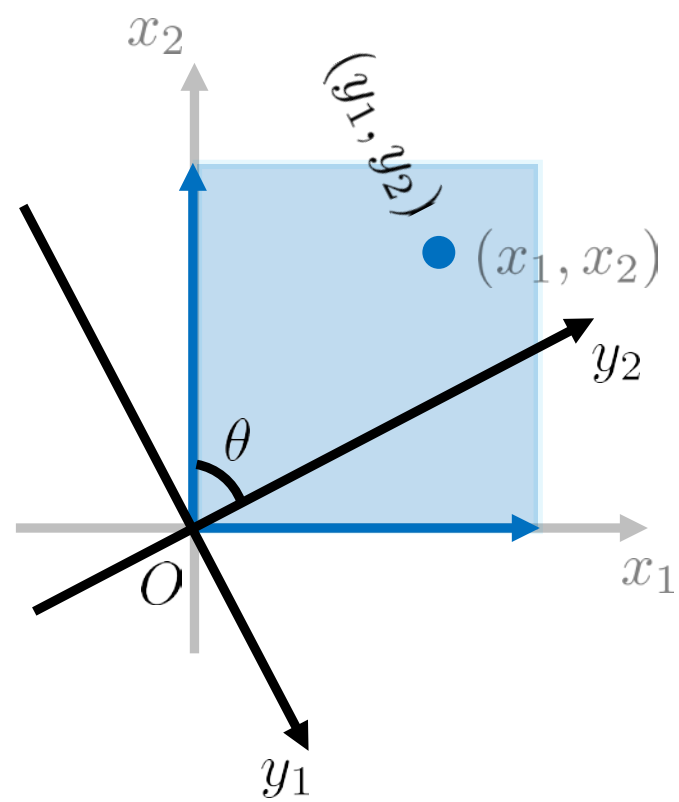
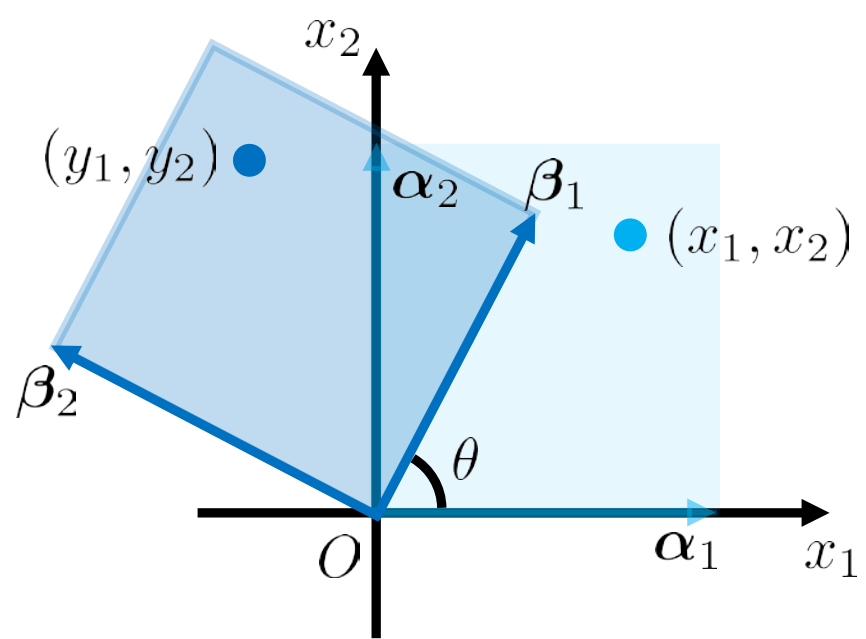
$$(5) \quad \begin{cases} y_1 = (1 + \cot \theta) \cdot x_1 + 0 \cdot x_2 \\ y_2 = 0 \cdot x_1 + 0 \cdot x_2 \end{cases}$$

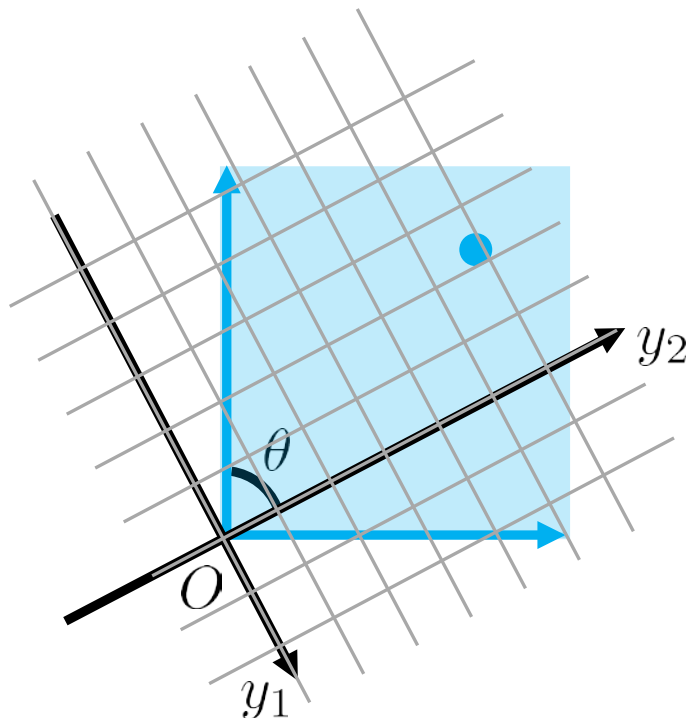
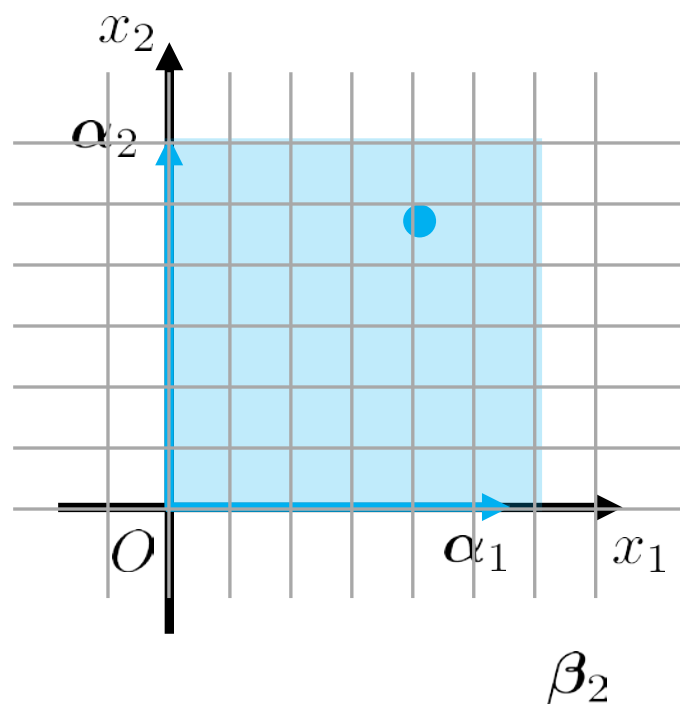




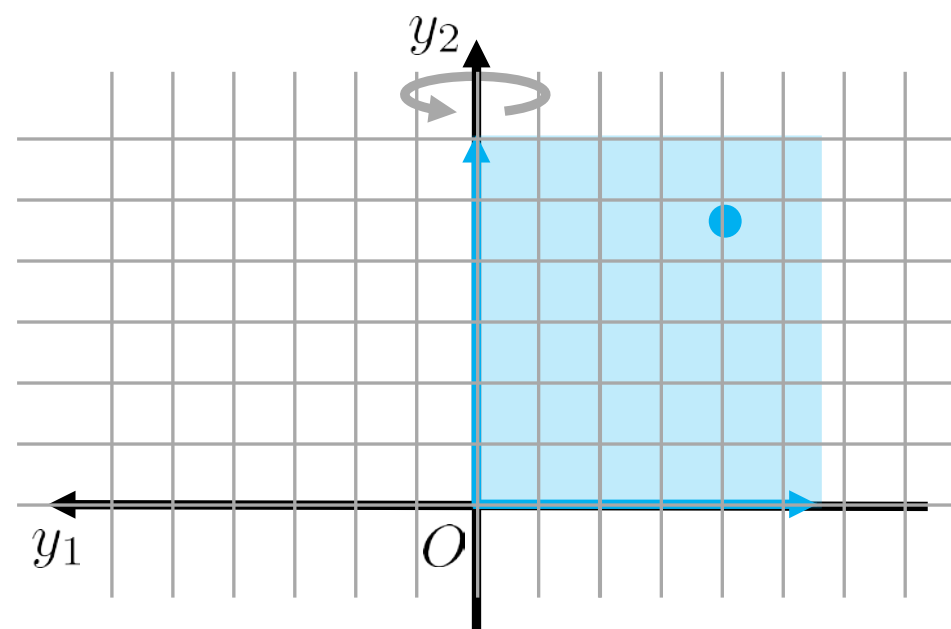
$$\begin{cases} y_1 = \cos \theta \cdot x_1 - \sin \theta \cdot x_2 \\ y_2 = \sin \theta \cdot x_1 + \cos \theta \cdot x_2 \end{cases} \iff \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\iff \mathbf{y} = A\mathbf{x} = (\beta_1 \ \beta_2)\mathbf{x}$$

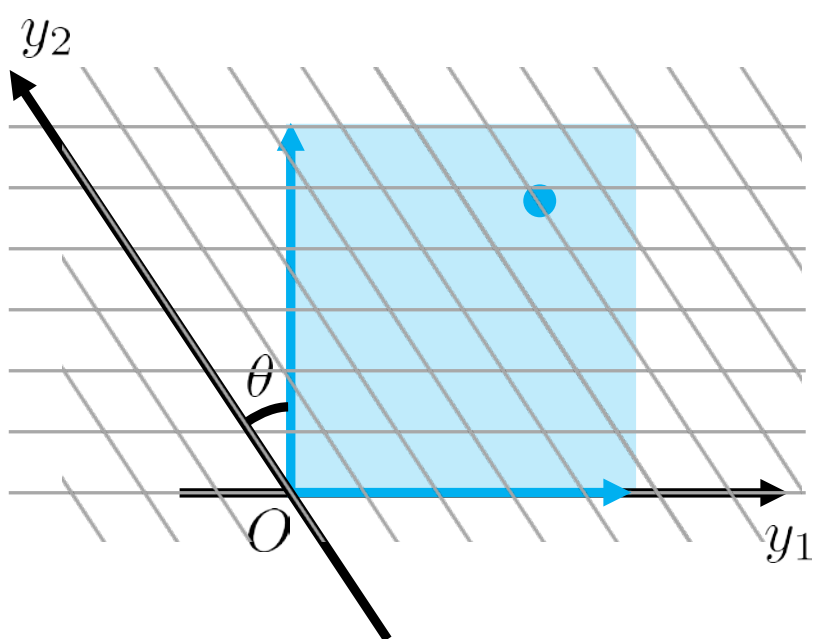




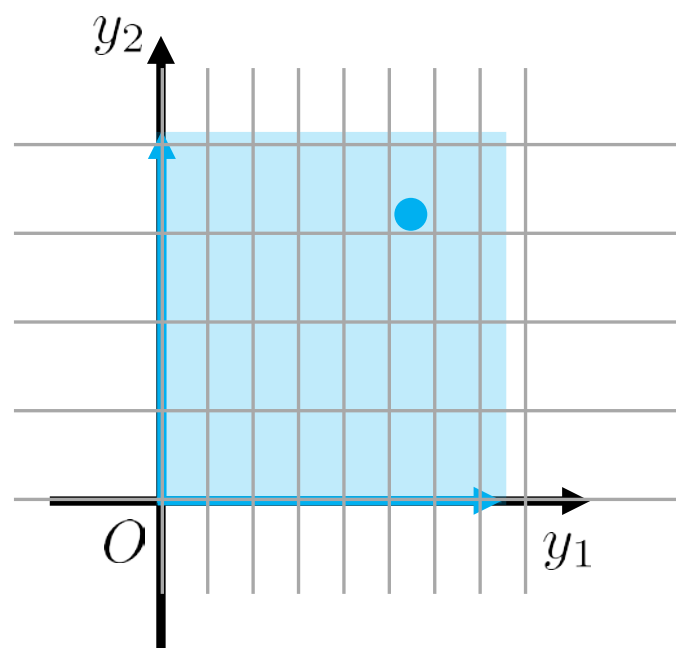
$$A_1 = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$



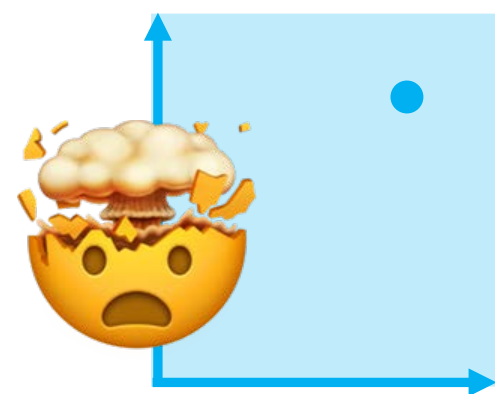
$$A_2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$



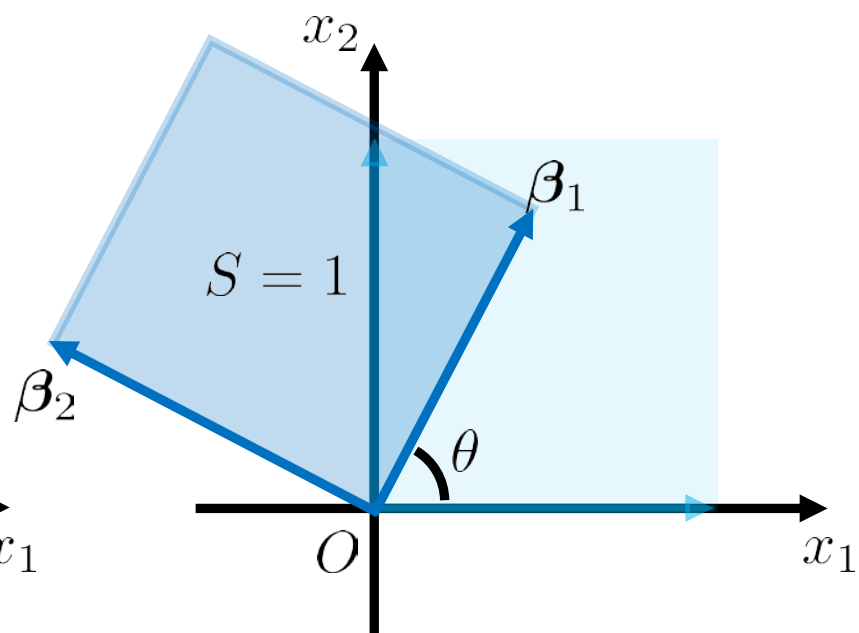
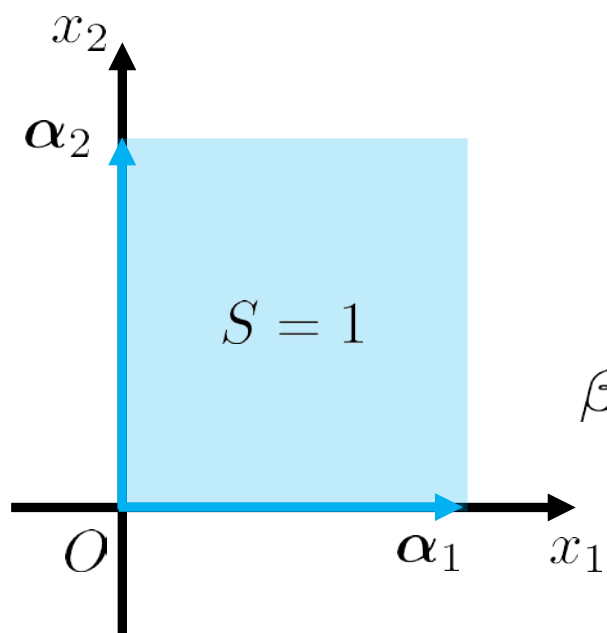
$$A_3 = \begin{pmatrix} 1 & \tan \theta \\ 0 & 1 \end{pmatrix}$$



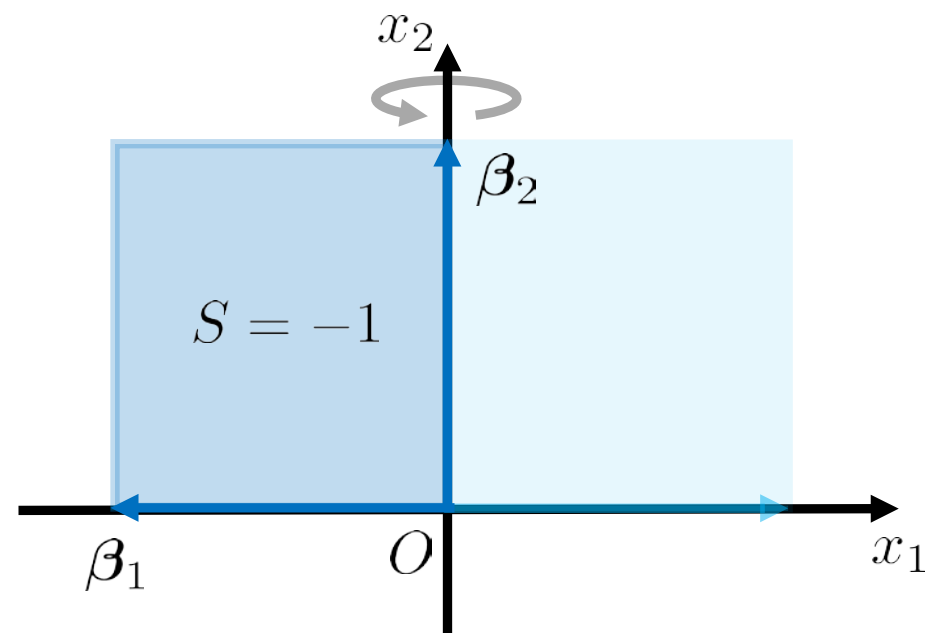
$$A_4 = \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix}$$



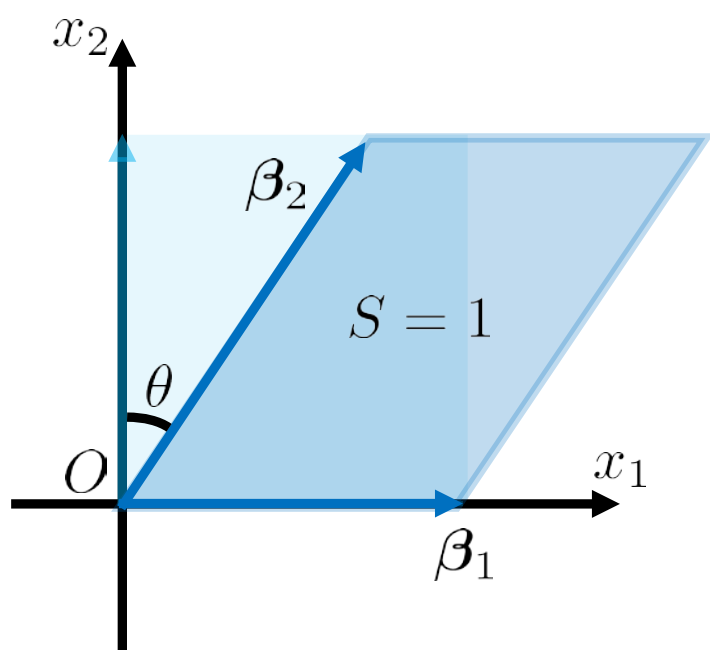
$$A_5 = \begin{pmatrix} 1 + \cot \theta & 0 \\ 0 & 0 \end{pmatrix}$$



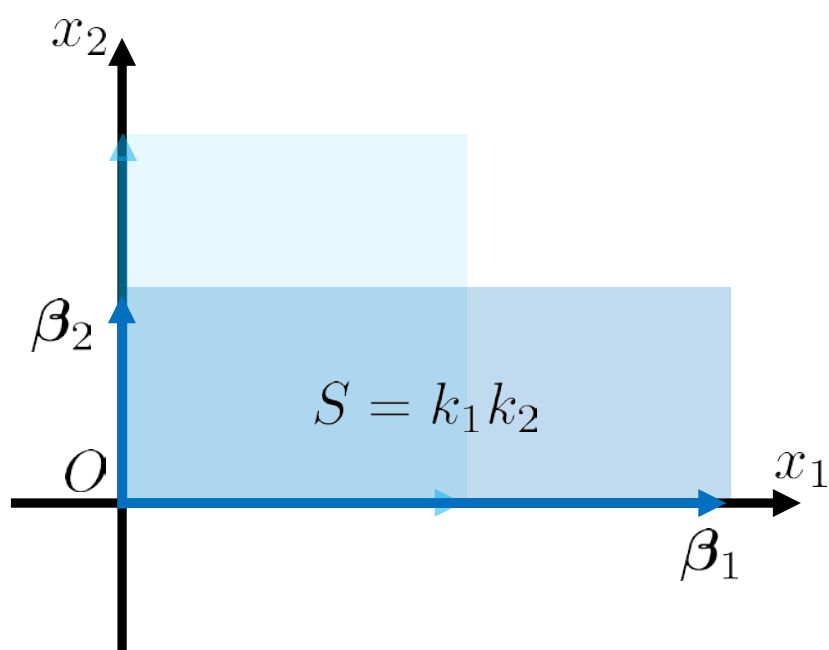
$$A_1 = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$



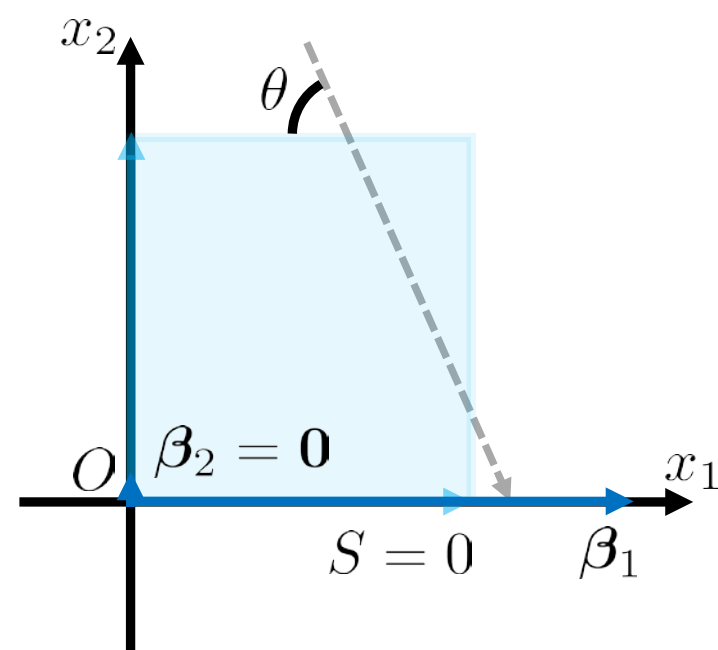
$$A_2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$



$$A_3 = \begin{pmatrix} 1 & \tan \theta \\ 0 & 1 \end{pmatrix}$$



$$A_4 = \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix}$$



$$A_5 = \begin{pmatrix} 1 + \cot \theta & 0 \\ 0 & 0 \end{pmatrix}$$